SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

Student Name	Pranav Chopade
PRN	202501110025
Division	A
Batch	A1
Roll No.	202501110025
Date of Submission	12-10-2026

2. Practical Details

Practical No.	4
Practical Title	DYNAMIC MEMORY USING REFERENCES, POINTERS, new AND delete
Date Performed	12-10-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

Write a C++ program demonstrating references and pointers. Allocate an integer array dynamically using the **new** operator, accept and display its elements using pointer notation, calculate their sum, access the first element using a reference, and release the allocated memory using **delete[]**.

4. Objectives

1. Understand references and pointers in C++.

2. Allocate memory dynamically using the new operator.

3. Access array elements through pointer notation and a reference.

4. Release dynamically allocated memory using delete[].

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Differentiate between pointers and references.

2. Allocate memory at runtime according to the required array size.

3. Traverse dynamically allocated arrays using pointers.

4. Deallocate memory safely after use.

6. Theory / Concept

Pointer: A pointer is a variable that stores the memory address of another variable.

Reference: A reference is an alias for an existing variable and can be used to access that variable.

Dynamic Memory Allocation: Memory can be allocated at runtime using the new operator.

new and delete: The new operator allocates memory; delete releases a single dynamically allocated object, while delete[] releases a dynamically allocated array.

7. Algorithm / Pseudocode

START

1. Read the number of elements from the user.
2. If the size is not positive, display an error and stop.
3. Allocate an integer array using new int[size].
4. Read the array elements.
5. Display elements using pointer notation *(arr + i).
6. Calculate and display the sum of all elements.
7. Create a reference to the first array element and display it.
8. Release the array using delete[] and set the pointer to nullptr.
9. Display a memory-release confirmation.

STOP

8. Technology / Tools / Libraries Used

Programming Language	C++
IDE / Platform	Dev C++
Libraries / Packages	iostream

9. Implementation / GitHub Repository

GitHub Repository URL	https://github.com/pranav-4797/OOP-Practical-Assignment
Repository Name	OOP-Practical-Assignment
Practical Folder	Practical 4
Branch Used	main
Commit ID / Latest Commit	Not provided

Repository Contents Used

The Practical 4 source file is provided separately as “Practical_4_Dynamic_Memory.cpp”. Add it to the Practical 4 folder of the repository if required. The output screenshot below uses sample input 5 and elements 10, 20, 30, 40, 50.

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
1	Size = 5; elements 10,20,30,40,50	Elements displayed; sum = 150	Sum = 150	Pass (sample run)
2	Size = 1; element 7	First element = 7; sum = 7	Expected from logic	To verify
3	Size = 0	Invalid size message	Expected from logic	To verify

11. Results / Observations

The program demonstrates dynamic allocation of an integer array, traversal using pointer arithmetic, reference-based access to the first element, calculation of the sum, and release of allocated memory using delete[]. For the sample input, the displayed elements are 10, 20, 30, 40, 50 and the sum is 150.

12. Source Code

```
#include <iostream>
using namespace std;

int main() {
int size;

cout << "Enter number of elements: ";
```

```
cin >> size;

if (size <= 0) {
cout << "Invalid size. Enter a positive number." << endl;
return 0;
}

int* arr = new int[size];

cout << "Enter " << size << " elements:" << endl;
for (int i = 0; i < size; i++) {
cin >> arr[i];
}

cout << "\nArray elements using pointer notation:" << endl;
for (int i = 0; i < size; i++) {
cout << *(arr + i) << " ";
}
cout << endl;

int sum = 0;
for (int i = 0; i < size; i++) {
sum += arr[i];
}

int& firstElement = arr[0];
cout << "First element using reference: " << firstElement << endl;
cout << "Sum of elements: " << sum << endl;

delete[] arr;
arr = nullptr;

cout << "Dynamic memory released successfully." << endl;
return 0;
}
```

Screenshots / Output

Practical 4 - Dynamic Memory - Output

```
Enter number of elements: 5
Enter 5 elements:
10 20 30 40 50

Array elements using pointer notation:
10 20 30 40 50
First element using reference: 10
Sum of elements: 150
Dynamic memory released successfully.
```

Sample input: size = 5; elements = 10, 20, 30, 40, 50.

13. Conclusion

The practical demonstrates how C++ pointers and references can be used with dynamically allocated memory. The array is created using new[], its elements are accessed through pointer notation, and the memory is released using delete[] to avoid leaving the allocation unreleased.

14. Assignment Questions

Q1. What is a pointer?
Answer: A pointer is a variable that stores the memory address of another variable.

Q2. What is the purpose of the new operator?
Answer: The new operator allocates memory dynamically at runtime and returns a pointer to the allocated memory.

Q3. What is the difference between delete and delete[]?
Answer: delete releases a single dynamically allocated object, whereas delete[] releases a dynamically allocated array.

Q4. What is a reference in C++?
Answer: A reference is another name, or alias, for an existing variable.

15. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	Pranav Chopade
PRN	202501110025
Signature	